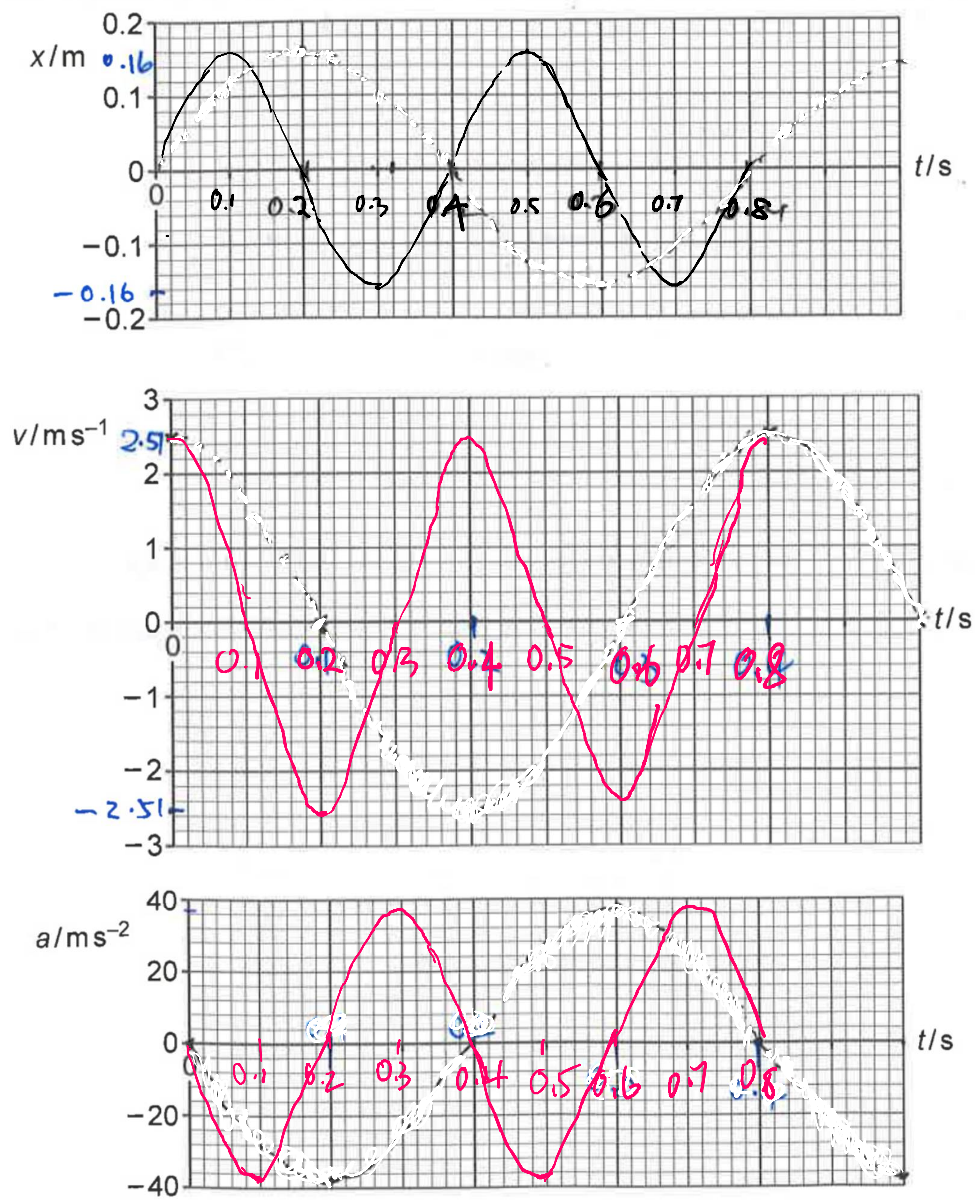
| | |
|---|---|
| | to S. The total energy is therefore smaller for P compared to S. |
| **3** | Comparing the given equation with a standard SHM equation $x = x_o \sin(\frac{2\pi}{T}t)$ gives<br>$x_o = 0.160\ m$<br>$15.7 = \frac{2\pi}{T}$<br>$T = 0.400\ s$<br>Max velocity magnitude = $0.16(15.7) = 2.51\ \text{m s}^{-1}$<br>Max acceleration magnitude= $15.7^2(0.16) = 39.4\ \text{m s}^{-2}$ |

| Qn | Solution |
|---|---|
| |  |
| 4(a) | The CG of a body is the single point at which the entire weight of the body can be |